

NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
First Semester B.Tech (CBCS) Degree Examinations

December 2023

CY1002-1 – CHEMISTRY FOR BIOMATERIALS
 (For BT)

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS

20 Marks

- The material which has the capacity to prevent mixing of active masses in the two components in a battery is.....
 A) Anode B) Cathode
 C) Separator D) Electrolyte
- In Li-ion battery, during discharging Li^+ flow through the membrane towards
 A) Cathode B) Anode
 C) Electrolyte D) None of these
- In the absence of a membrane in methanol-oxygen fuel cell, methanol diffuses through the electrolyte into the cathode and undergoes
 A) Reduction B) Oxidation
 C) Degradation D) Substitution.
- Permanent hardness of water is caused due to the presence of.....
 A) MgSO_4 B) $\text{Ca}(\text{HCO}_3)_2$
 C) CaCO_3 D) $\text{Mg}(\text{HCO}_3)_2$
-is the indicator used in the determination of Dissolved oxygen by Winkler's method.
 A) Eriochrome black-T B) Starch
 C) Phenolphthalein D) Diphenylamine
- Anion exchange resins can be regenerated by giving wash.
 A) Alcohol B) Acid
 C) Alkali D) Water
- Secondary wastewater treatment is also called as
 A) Lime-soda process B) Chemical treatment
 C) Physical treatment D) Biological treatment
- The difference in potential which causes a current to flow from an electrode of higher potential to lower is called.....
 A) Reduction potential B) Oxidation potential
 C) Electromotive force D) Standard electrode potential
- Which of the following is used to determine pH of unknown solution?
 A) Platinum electrode B) Glass electrode
 C) Standard hydrogen electrode D) Gas electrode
- Conductometer consists of two and a conductance measuring device.
 A) Platinum electrode B) Calomel electrodes
 C) Standard hydrogen electrodes D) Glass electrodes
- Tg of Polyethylene is lower than Polypropylene because
 A) Stiff backbone B) Higher molecular mass
 C) Strong polar attraction between the molecules D) Flexible backbone

12. **Functionality of Phenol is**
 A) One B) Two
 C) Three D) Four
13. **In thermosetting resins, all monomer units are held together through.....bonds**
 A) Primary covalent B) Secondary cohesive
 C) Ionic D) Hydrogen
14. **The percentage of hexadecane present in a mixture of hexadecane and 2-methyl naphthalene is called.....**
 A) Octane number B) Cetane number
 C) Transport number D) None of the above
15. **The IUPAC name of the compound**
- $$\begin{array}{c} \text{CH}_3\text{Cl} \\ | \\ \text{CH}_3-\text{CH}_2-\text{CH}-\text{CH}-\text{CH}_3 \end{array}$$
- A) 3-Methyl-4-chloro pentane B) 2-Chloro-3-methyl pentane
 C) 3-Methyl-4-chloro hexane D) 2-Chloro-3-ethyl pentane
16. **.....are present in every living organism to carry out a wide range of chemical reactions, also finds application as an important tools used in Green chemistry.**
 A) Enzymes B) Chemical catalysts
 C) Carbohydrates D) Proteins
17. **Adipic acid prepared fromis the green route of synthesis**
 A) Hexane B) Benzene
 C) Hexene D) Acetic acid
18. **Materials having one dimension in nano scale are called nanomaterials.**
 A) Zero Dimensional B) Two Dimensional
 C) One Dimensional D) Three Dimensional
19. **ZnO nanoparticles were synthesized by direct precipitation method using as precursors.**
 A) Potassium nitrate and NaOH B) Sodium nitrate and KOH
 C) Zinc nitrate and KOH D) Zinc nitrate and KCl
20. **In nanomaterials the melting temperaturewith the decrease in the particle size diameter.**
 A) Decreases B) Increases
 C) Remains same D) None of the above

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1.	a) Define a battery. Explain the construction and working of Na-ion battery.	7	L2	1	1
	b) Describe the construction, working and applications of CH ₃ OH-O ₂ fuel cell.	6	L2	1	1
	c) Distinguish between fuel cell and battery. Mention the advantages of fuel cell.	3	L4	1	1
2.	a) Explain the Winkler's experimental method of determination of dissolved oxygen.	6	L2	1	1
	b) Describe the softening of water by hot lime-soda process.	7	L2	1	1
	c) In a COD experiment, 28.5 cm ³ and 19 cm ³ of 0.025N FAS solution were required for blank and sample titration respectively. The volume of test sample used is 25 cm ³ . Calculate the COD of the sample solution.	3	L3	1	2
3.	a) Explain the construction, working and applications of Photovoltaic cell.	6	L2	1	1
	b) Discuss any three causes of scale formation in boiler.	3	L1	1	1
	c) Explain the following (i) Reverse osmosis method of desalination of water (ii) Activated sludge process of sewage treatment	7	L2	1	1

Unit – II

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|-------|---|---|----|---|---|
| 4. a) | What is concentration cell? A concentration cell was constructed by immersing two copper electrodes in 0.01M and 10 M CuSO ₄ solution. Write the cell scheme, cell reactions and calculate the e.m.f. of the concentration cell. | 5 | L3 | 2 | 2 |
| b) | Give the construction of glass electrode. Explain the experimental method of determination of pH of a unknown solution using a glass electrode. | 7 | L2 | 2 | 1 |
| c) | Explain the experimental method of Conductometric titration. | 4 | L2 | 2 | 1 |
| 5. a) | Explain the free radical mechanism of addition polymerization taking vinyl chloride as an example. | 6 | L2 | 2 | 1 |
| b) | What is polymer composite? Explain the synthesis and applications of carbon fibre. | 5 | L2 | 2 | 1 |
| c) | Explain the mechanism of oxidative doping of polyacetylene. | 5 | L2 | 2 | 1 |
| 6. a) | Describe the functions of different parts of Biosensor with neat diagram. | 6 | L2 | 2 | 1 |
| b) | Define glass transition temperature. Explain the any four factors affecting the T _g . | 5 | L2 | 2 | 1 |
| c) | Explain the mechanism of knocking in diesel engine | 5 | L2 | 2 | 1 |

Unit – III

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|-------|---|---|----|---|---|
| 7. a) | List out the sources and ill effects of NO _x pollution | 4 | L1 | 3 | 1 |
| b) | Discuss any six principles of Green Chemistry | 6 | L2 | 3 | 1 |
| c) | Describe the synthesis of nanomaterials by Chemical vapour deposition method. | 6 | L2 | 3 | 1 |
| 8. a) | Write a note on solvent free organic synthesis. | 5 | L2 | 3 | 1 |
| b) | Explain the synthesis of conventional and green route synthesis of Adipic acid. | 5 | L2 | 3 | 1 |
| c) | Explain the synthesis of nanomaterials by Sol-gel method. | 6 | L2 | 3 | 1 |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
